1 Background

The chapter that my problem comes from is chapter 12 about nonlinear mechanics and chaos. This section of mechanics is important because there are many systems that are nonlinear and even if a system obeys equations of motion, its future behavior may be unpredictable. In order for a system to exhibit chaos, its equations of motion must be nonlinear. For example, $\ddot{\theta} = \cos(\theta)$ would be a nonlinear differential equation for θ because $\sin(\theta)$ is nonlinear in θ. Before this chapter, when we had nonlinear problems we used approximations that made them linear but that is not something we can do for all problems. It is important to note that if a system has nonlinear equations of motions, it is not guaranteed to be a chaotic system, for example a simple pendulum never exhibits chaos. A distinction for a chaotic system is a sensitivity to initial conditions. In a chaotic system, objects that start near each other end up far from each other. This is one of the ways that we will determine if the system we are dealing with is chaotic throughout this problem. Additionally, most nonlinear equations are not able to be solved analytically so I will solve this problem numerically using python.

My problem is The Swinging Atwood Machine. It is similar to the Atwood Machine in chapter 7 but has an additional level of complexity because one of the masses is free to move in 2-dimensions rather than just one. Essentially, it is a pendulum in which the length is able to change. A visual of the system can be seen in Figure 1. Unlike the normal Atwood machine, an Atwood machine in which one of its masses can oscillate can lead to some highly interesting non-linear behaviour.

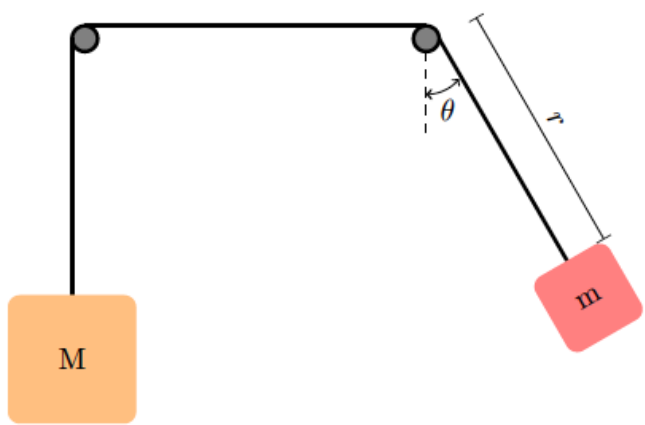

Figure 1: This figure shows The Swinging Atwood. It is similar the Atwood Machine we saw in chapter 7 but one of the masses in free to move in two dimensions rather than just one.

2 Solving the Problem

For the Swinging Atwood Machine I was given that the system had a kinetic energy of

$$T = \frac{1}{2}M\dot{r}^2 + \frac{1}{2}m(\dot{r}^2 + r^2\dot{\phi}^2) \tag{1}$$

and a potential energy of

$$U = Mgr - mgr\cos(\theta) \tag{2}$$

2.1 Part a.)

In this part, we are solving for the equations of motion of $\ddot{r}$ and $\ddot{\theta}$. To do this we will use the Lagrangian methods that were used in chapter 7, except we will not be making any simplifications to make the system linear. We start by finding the Lagrange, which is $L = T - U$. I found this to be

$$L = -Mgr + \frac{1}{2}M\dot{r}^2 + gmr\cos(\theta) + \frac{1}{2}m(\dot{r}^2 + r^2\dot{\phi}^2). \tag{3}$$

We solved the Lagrange equation in python. Figure 2 shows the code that was used to solve the Lagrangian for θ. Once I solved the Lagrangian I solved for $\ddot{\theta}$ which is the equation of motion of $\ddot{theta}$. The equation of motion is

$$\ddot{\theta} = -\frac{g\sin(\theta) + 2\dot{r}\dot{\theta}}{r}. \tag{4}$$

```
# Lagrange for θ

LHS_θ = L.diff(θ)
RHS_θ = L.diff(θdot).diff(t)
Lag_θ = LHS_θ - RHS_θ
motion_θ = sym.solve(Lag_θ,θdoubledot)
motion_θ[0]
```

Figure 2: This is the code that was used to find the equation of motion for $\ddot{\theta}$.

We did the same thing for r. The code to find the equation of motion of $\ddot{r}$ is shown in Figure 3. The equation of motion is

$$\ddot{r} = \frac{-Mg + gm\cos(\theta) + mr\dot{\theta}^2}{M + m}. \tag{5}$$

```
# Lagrange for r
LHS_r = L.diff(r)
RHS_r = L.diff(rdot).diff(t)
Lag_r = LHS_r - RHS_r
motionr = sym.solve(Lag_r, rdoubledot)
motion_r = motionr[0]
motion_r
```

Figure 3: This is the code that was used to find the equation of motion for $\ddot{\theta}$.

2.2 Part b.)

In order to simplify equation 6, we can let $\mu = \frac{M}{m}$.

$$\frac{-Mg + gm\cos(\theta) + mr\dot{\theta}^2}{M + m}$$

$$= \frac{-Mg + gm\cos(\theta) + mr\dot{\theta}^2}{m(M/m + 1)}$$

Now we can substitute μ for $\frac{M}{m}$ to get

$$\frac{-Mg + gm\cos(\theta) + mr\dot{\theta}^2}{m(\mu+1)}.$$

Now if we simplify we can rewrite the equation of motion for r as

$$\ddot{r} = \frac{-\mu g + g\cos(\theta) + r\dot{\theta}^2}{(\mu+1)}. \tag{6}$$

2.3 Part c.)

In this part we will find if there are any equilibrium points. We know at equilibrium the system is stationary so $\dot{r} = \dot{\phi} = \ddot{r} = \ddot{\phi} = 0$. We can plug these into each equation and solve each of the equations to find the equilibrium points.
If we start with finding the equilibrium point for θ

$$0 = -\frac{g\sin(\theta)}{r} \tag{7}$$

we solve for θ and conclude that the system is at equilibrium when $\theta = 0$.
We repeat the same process for the r equation of motion

$$0 = \frac{-Mg + gm\cos(\theta)}{M+m}. \tag{8}$$

This equation doesn't have a dependence on r, but we can conclude that equilibrium will exist when $M = m$.

2.4 Part d.)

Here is where we start to visualize the trajectory of the swinging mass. In this part we are going to let $\mu = 7.7$, $r_0 = 1$, $\theta_0 = \pi\ 2$, $\dot{r} = 0$ and $\dot{\theta} = 0$. As mentioned in the background I will solve for the trajectories numerically using the Euler-Cromer method. The code can be seen in Figure 4. We plotted the trajectory in a polar plot that can be seen in Figure 5. We can see that the mass follows the same path each cycle indicating the motion is periodic and not chaotic. To confirm this, we repeated the process again with $r = 1.005$ and determined the distance between the two objects at each point in time. We plotted the distance over time. This distance only gets larger because of the steady increase in the amount of error overtime due to the periodic nature. This system is not chaotic.

```python
g = 9.81 #m/s^2 --> gravity
θ = np.pi/2 #radians
r = 1 #initial length of the pendulum potion
vr = 0 #initial velocity in the r direction
vθ = 0 # initial velocity in the θ direction
μ = 7.7
t = 0 #second --> Initial time
dt = .0001 #second ---> Time step

# defining empty lists that we can save our data in
rs =[]
θs = []
ts = []
while t <= 5:
    # Calculate the acceleration
    ar = (-μ*g + g*np.cos(θ) + r*vθ**2)/(μ + 1)
    aθ = -(g*np.sin(θ) + 2*vr*vθ)/ r
    #Update Velocity
    vr = vr + ar*dt
    vθ = vθ + aθ*dt
    # Update position
    r = r + vr*dt
    θ = θ + vθ*dt
    # Append these to our lists
    rs.append(r)
    θs.append(θ)
    # Update time
    t = t + dt
    ts.append(t)
```

Figure 4: This is the code that was used to numerically solve for the trajectory of the swinging mass. In order to change this code for each part of this problem, we only need to change the initial conditions and μ value.

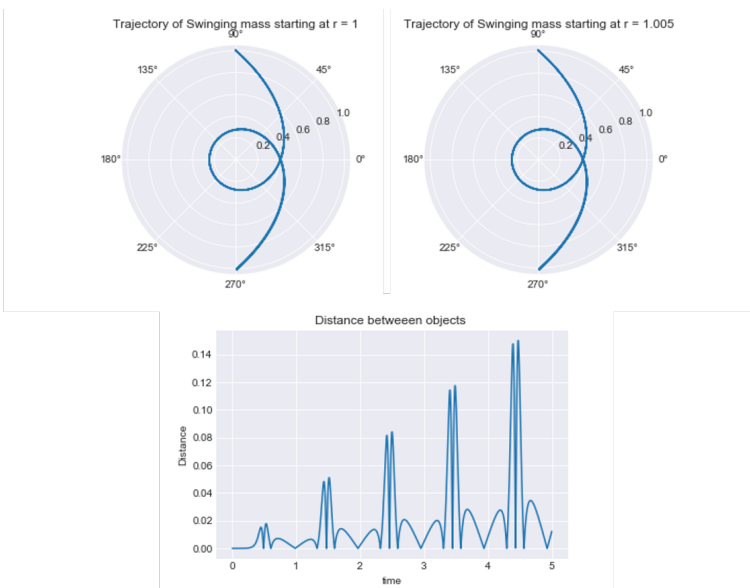

Figure 5: The top plots are of the trajectory of the swinging mass at two different r_0 values and $\mu = 7.7$. The bottom plot is the distance between the two masses. We can see that the mass follows the same path with each cycle.

2.5 Part e.)

In this part we followed the exact same steps that we followed above and used the code shown in Figure 4 except we used $\mu = 3$ and ran it for a longer period of time. The polar plots of the two different r values and the distance between the two objects are shown in figure 6. Although this motion looks more complex than the previous case it is still not chaotic, rather it is quasi-periodic as it doesn't exactly repeat itself exactly each cycle.

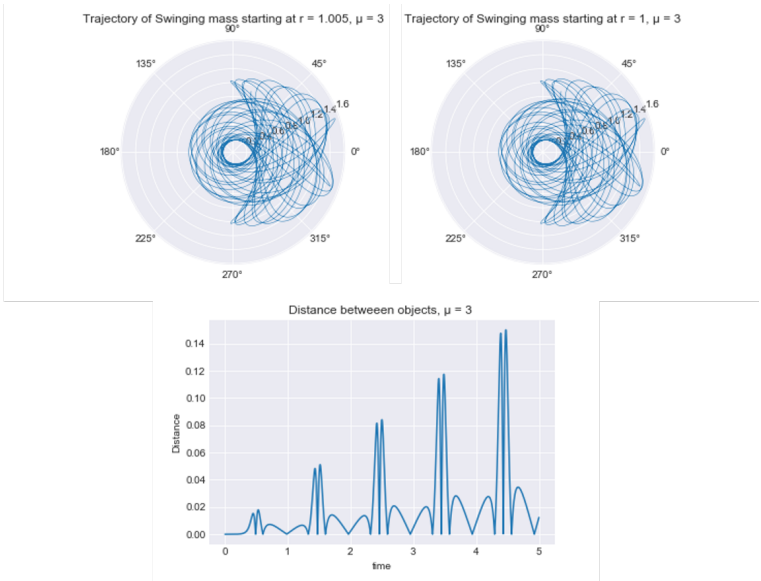

Figure 6: The top plots are of the trajectory of the swinging mass at two different r_0 values and $\mu = 3$. The bottom plot is the distance between the two masses. This motion is quasi-periodic.

2.6 Extra fun part

We added this part in because we hadn't seen any example of chaos yet. Again we used similar code to Figure 4 but used $\mu = 10.63$. When we plotted the trajectory it definitely looked more chaotic than the previous two as we can see in Figure 7. Additionally, when we plotted the distance between the two objects we got a graph in which the distance was immediately over 2 and not steadily increasing due to periodic error. This confirms that this motion is chaotic.

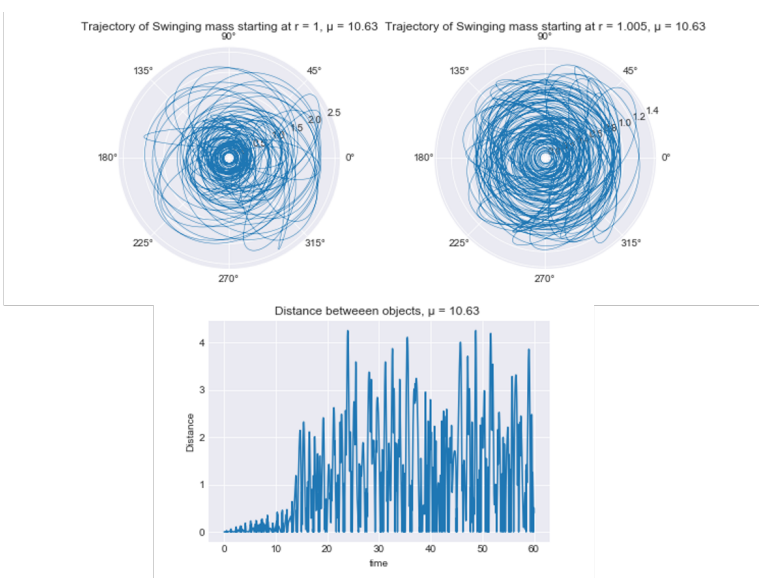

Figure 7: The top plots are of the trajectory of the swinging mass at two different r_0 values and $\mu = 3$. The bottom plot is the distance between the two masses. This motion is chaotic.

2.7 Part f.)

In this final part we started with initial conditions $r_0 = 1, \theta_0 = \pi\ 3, \dot{r} = 0, \dot{\theta} = 0$. The goal is to find a μ value between 1 and 5 that corresponds to a periodic orbit. To do this I used the same Euler-Cromer method as in Figure 4 but plugged in our new initial conditions. I then guess and checked to find a μ that corresponded to periodic behavior. I ended up finding that $\mu = 2.811$ gives us periodic behavior. We know that it is periodic because it traces over the same path every cycle as seen in Figure 8. Just to be sure it traced over the same path, I plotted the trajectory for 60 seconds.

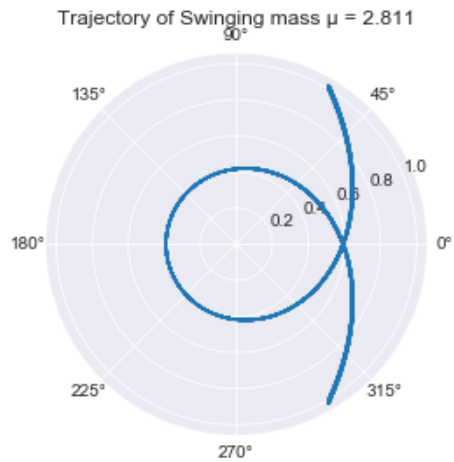

Figure 8: The top plots are of the trajectory of the swinging mass at two different r_0 values and $\mu = 3$. The bottom plot is the distance between the two masses. This system is periodic

3 Conclusion

In this project we looked at the trajectory of the two masses in the Swinging Atwood Machine. We are able to see from the above parts that the μ value and the initial conditions determined if this system was chaotic or not.